

Fateh Ali Syed Bukhari

(587) 664-1333 | ShahjiFateh@gmail.com | linkedin.com/in/fatehalibukhari | github.com/FatehPHD | fatehali.vercel.app

Education

University of Calgary

Calgary, AB

Bachelor of Science, Schulich School of Engineering — Major: Software Engineering

Sep 2022 – Present

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management, Software Architecture, Object-Oriented Design, Computer Networks, Distributed Systems, Machine Learning

Technical Skills

Languages: Java, Python, JavaScript, C/C++, SQL, HTML/CSS

Frameworks & Libraries: React, Node.js, Spring Boot, Flask, pandas, NumPy

AI & APIs: GPT-4o / OpenAI API, REST APIs, FFmpeg, OCR

Cloud & DevOps: Google Cloud Platform, Firebase, Docker, Git

Databases: MySQL, PostgreSQL, Supabase

Experience

Backend Lead

Jul 2025 – Present

Enactus Calgary

- Led a 4-person backend team building a financial literacy platform now used by 200+ high school students across 2 schools.
- Eliminated auth-related support tickets by designing a hybrid Firebase Auth + MySQL system on GCP, securing data for 200+ users.
- Cut buffering complaints by 35% by building an FFmpeg-powered adaptive streaming pipeline delivering smooth playback across all device types and network speeds.
- Reduced average API response time by 30% through optimized query design while engineering REST APIs for quizzes, leaderboards, and course modules backed by Google Cloud Storage.

Projects

Study Planner | React, Flask, Supabase, GPT-4o, PostgreSQL | [Demo](#) | [GitHub](#)

Jul 2025

- Cut student calendar-setup time from 40 minutes to seconds by building an AI planner that OCR-parses course outlines into 50+ auto-generated deadlines and personalized schedules.
- Integrated GPT-4o for natural-language schedule optimization across PDF, Word, and plain-text inputs, supporting varied student workflows.
- Architected a full-stack platform with real-time Supabase sync and Row-Level Security, isolating each student's data at the database layer.
- Shipped dashboards, calendar views, what-if grade calculators, and admin analytics with a 4-person team in a single semester.

Movie Theatre Booking System | Java, C++, React, Spring Boot, MySQL, Git | [GitHub](#)

Dec 2024

- Engineered an end-to-end ticket booking system handling seat selection, secure payment processing, and automated email confirmations.
- Prevented double-bookings under concurrent load using row-level locking and transactional writes across 25+ simultaneous sessions.
- Implemented role-based access control across guest, registered, and admin tiers with differentiated pricing and feature access.
- Designed a normalized relational schema managing 1,000+ user profiles, payment records, and concurrent ticketing transactions without data conflicts.

Subway Information Display | Java, REST APIs, MySQL, Java Swing | [GitHub](#)

Jul 2024

- Reduced average rider query time by 30% through automated announcements and a proactive train-tracker overlay.
- Rendered 100+ real-time train updates, station info, and transfer connections with sub-second refresh on a live transit display.
- Built a MySQL-backed ad rotation engine scheduling 50+ ads across configurable time slots and display formats.
- Integrated external weather and news APIs, applying relevance filtering across 200+ results to surface only contextual content.

Arcade Joystick Game | C++, Arduino | [GitHub](#)

May 2021

- Built an arcade-style joystick game on Arduino in C++ with an adaptive difficulty system that shrinks obstacle gaps and increases speed as the score rises.
- Tuned difficulty curves across 200+ test rounds to sustain engagement while optimizing collision detection, input handling, and scoring logic for smooth real-time gameplay.
- Implemented collectible stars and power-ups, a persistent high-score system, and a rules/options menu on constrained embedded hardware.